Master plan after the scout (2026-10-20)

Status: planning document. Everything marked "exploratory" was run once on a development half; nothing here is a pre-registered result. Sources: four literature/dataset scouts (`review/scout/SCOUT_1` to `SCOUT_4`; they ran on Sonnet because the `opus` model alias is unavailable in this environment and Fable needs usage credits), my own checks of their key citations on arXiv/GitHub/Hugging Face, and one new measurement on a public natural-error dataset.

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1. Executive summary: what changed

1. **A first measurement on natural (model-made) errors is bad news for the deterministic checker.** On the development half of the public DeRef data (Ma et al., arXiv 2610.18497; BFCL-based `fcrb` and ComplexFuncBench-based `cfb`), the type-aware argument-grounding score separates wrong calls from correct calls at roughly chance level: call-level AUROC **0.48 (fcrb) and 0.57 (cfb)**, with 51% and 42% of *correct* calls blocked at our default threshold (after fixing a number-matching bug found while investigating; see Section 3). Our headline "7.3% false blocks vs 14.0% for exact match at 96.5% detection" is true only for **injected absent-value substitutions in LLM-generated Glaive conversations**; it does not transfer to natural errors. (Section 3.)
2. **Why:** in these datasets many wrong values are *visible but wrong choices* (a grounding check cannot see this), and many correct values are *not stated in the text* (translated, defaulted, inferred), so "unsupported" does not mean "wrong". The scouts' count: only 54% of wrong arguments in `fcrb` are absent from context, and "unsupported" is wrong in only 44% (fcrb) and 62% (cfb) of cases. This is a ceiling on **any** grounding check, including LLM judges that rely on visible text.
3. **The professor's concern is real and the closest paper already answers part of it against us.** Ma et al. report plain-input judge argument AUROC of 72.5 to 91.95 on fcrb, 63.4 to 78.9 on cfb, and 83.4 to 89.3 on their rule-injected FinTrace set (their Table 2, as read by Scout 1). So strong judges are a serious baseline, and we must expect them to beat a purely deterministic checker on natural errors.
4. **Several ingredients are not ours to claim:** the weakest-argument call score (their `min` aggregation), traceability of a call to its context (ProvenanceGuard), per-argument provenance and origin tracking (PACT, ROPE, SARA, ToolFence, Agent-Sentry, SIEVE, FIDES), "deterministic = immune to injection" (ROPE's thesis), deterministic fuzzy argument grounding as software (the small `Saidso` repository, README only), split-conformal false-alarm budgets for verifiers (E-valuator, CORA, ConformalGuard, role-stratified CRC).
5. **What remains defensible and worth doing** (Section 4): (a) a rigorous measurement of **where grounding checks work and where they cannot**, per derivation path, for deterministic checkers and LLM judges on the same data; (b) a **certified cascade** that uses the deterministic stage only where it is reliable and an LLM for derived values, with an end-to-end false-block guarantee; (c) a correctly stated, calibration-conditional guarantee; (d) a **head-to-head under injected text**, which nobody has run against a deterministic grounding check; (e) the benchmark artefacts and negative results.
6. **Recommended repositioning (decision for you in Section 13):** move from "we built a hallucination detector" to **"an evaluation-first study of argument grounding for tool calls, with a calibrated cascade as the engineering outcome"**. This is the honest answer to "if LLMs beat this there is no use": we measure exactly where they do, where nothing grounded can work, and what a cheap deterministic stage is still good for.
7. **Our own earlier wording needs corrections** (Section 11): the guarantee text, the novelty claims, and scoping of the 7.3% vs 14.0% result.

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2. Scout findings, verified state

I verified by fetching arXiv abstract pages / GitHub / Hugging Face APIs that every item in the following table exists with the stated title, date and licence. "Read" is what the scout read; I did not re-read papers beyond abstracts. Treat numbers as the authors' unreplicated claims.

Finding	Source (verified to exist)	What it means for us
Argument-level LLM judges on injected and natural single-argument errors; judges accept correct values less often when no text states them (by 22.7 to 36.1 points), prefer a visible wrong value to a hidden correct one; fix = candidate-blind derived reference (DeRef); plain-input judge argument AUROC 63 to 92 depending on dataset and judge	Ma et al., arXiv 2610.18497 (13 Oct 2026); data HF `lm2445/DeRef` (Apache-2.0), code `SarielMa/DeRef_release` (Apache-2.0)	Closest prior art and the best test set. Their weakness finding also applies to us. Their threshold rule has no guarantee.
Pre-execution traceability to context, all-LLM pipeline, about 3.6 calls and 2.9 to 7.5 s per case	ProvenanceGuard, arXiv 2607.01236	"Traceable to context" is not our novelty. Add its prompt style as a baseline.
Argument-level provenance and origin enforcement, deterministic monitors	PACT 2605.11039, ROPE 2608.27496, SARA 2608.27146, ToolFence 2609.37196, Agent-Sentry 2603.22868, SIEVE 2512.06716	Not novel. Open gap they all name: *deriving* provenance without an LLM or agent self-report; ROPE's exact matching refuses reworded values.
Deterministic fuzzy argument grounding with tool-output provenance, MIT, 1 star, no evaluation (README read only)	`KarthikRommula/Saidso`	Closest *software*. Install and run it as a baseline on our data.
Conformal / false-alarm control for verifier scores	E-valuator 2512.03109, CORA 2604.09155, ConformalGuard 2610.14365, role-stratified CRC for tool calls 2607.24343	Calibration is established; applying it to argument grounding is a modest extension, not a major claim.
Evidence a deterministic layer fails *independently* of LLM judges, and that two LLM judges' misses correlate	Yang, arXiv 2610.07359 (shell/cloud actions, single author)	Gives the independence measurement to copy (phi coefficient, multiplication-equivalent layers).
Evidence source matters more than verifier-model diversity	VP-CONTROL 2609.10969	Our checker reads the same conversation as the agent; independence is **not** automatic and must be measured.
LLM judges / guard models can be manipulated by text in context	2610.12292 (typed decision models), 2504.18333, adaptive attacks 2503.00061 and 2510.09023, 2504.11168, 2610.10742	Motivates the robustness hypothesis. Counter-evidence exists (small effects for some judges), so we test it.
Grounding "in context" is itself attackable: one planted line flips a naive provenance check 94 to 99%	2610.09793	Our checker needs a source-trust label before any injection claim.
Conformal facts: realised false-block rate for one calibration set is Beta-distributed; about half of calibration draws exceed alpha; high-probability version needs the binomial order-statistic rule; cascades should be certified end to end (Learn-then-Test); union-bound loss for tier-by-tier calibration	Angelopoulos and Bates 2107.07511, Bates et al. 2104.08279, conformal risk control 2208.02814, Learn-then-Test 2110.01052, Trust or Escalate 2407.18370, Conformal Cascade 2607.25018, Cheap Verifiers 2609.01345	Corrects how we have described our guarantee; defines the cascade statistics.
Public datasets: DeRef (natural errors, per-argument labels with derivation paths), per-model raw BFCL outputs (109 models), tau-bench trajectories (MIT), AgentHallu BFCL human labels (36 argument cases), AgentDojo runs (MIT)	HF `lm2445/DeRef`; GitHub `HuanzhiMao/BFCL-Result` (Apache-2.0, 3.1 GB); `sierra-research/tau-bench` (MIT); AgentDojo `runs/`	We do **not** need to run agent models to get natural errors. The Hallucinated-Tools Benchmark has no findable download.

Gaps the scouts could not close: older dialogue-state-tracking prior art, several cited-only papers (TwinCheck, ToolPRMBench, CLM blog), the DeRef code was not read, no verified reference for conformal prediction under

conversation-level dependence, classical non-inferiority references only as metadata. Search coverage was arXiv-centric; absence of a competitor is not proof of absence.

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3. First evidence on natural errors (exploratory, dev half only)

Script `benchmarks/explore_deref_dev.py`, data downloaded from `lm2445/DeRef` (3.7 MB + 28.5 MB). Split by `query_id`, seed 11; **only the dev half was read**; no threshold tuned. Context = request + all history text. Labels and derivation paths are the dataset authors' own automatic labels.

Argument level: share flagged UNGROUNDED by our matcher

Dataset	Derivation path	Correct	Wrong
fcrb	visible	0.3% (n=383)	0.0% (n=139)
fcrb	no_source	65.6% (n=256)	68.2% (n=201)
fcrb	default	69.4% (n=98)	89.3% (n=28)
cfb	visible	18.9% (n=438)	15.6% (n=96)
cfb	history_only	1.4% (n=138)	0.0% (n=22)
cfb	no_source	82.7% (n=254)	61.9% (n=433)

Call level (fcrb dev: 315 correct + 315 wrong; cfb dev: 174 correct + 333 wrong)

Dataset	AUROC of (1 - call score)	Exact match: correct blocked / wrong blocked	Type-aware (tau 0.88): correct blocked / wrong blocked
fcrb	0.482	67.9% / 58.1%	51.7% / 44.4%
cfb	0.560	49.4% / 78.7%	43.1% / 49.5%

Reading it honestly.

- A grounding score cannot detect a *visible-but-wrong* value (fcrb: 139 wrong arguments, 0% flagged). That is by construction.
- For values with no textual source, correct and wrong are flagged at similar rates (fcrb 66% vs 68%; cfb even reversed, 83% vs 62%), so lexical grounding has no discriminating power there. Reasons visible in the first example: a Korean string for an English request (translation), defaults, inferred values.
- **Bug found and fixed:** 19% of cfb "visible" correct arguments were flagged because our number matcher ignored minus signs (all 83 flagged values were negative numbers, mostly longitudes and latitudes). After the fix 0% are flagged and the conclusion is unchanged: fcrb AUROC 0.479 (51.4% of correct calls blocked, 44.1% of wrong), cfb AUROC 0.570 (42.0% / 48.6%). The numbers in the two tables above are the pre-fix ones; the post-fix values are these. The fuzzy matcher was also 45 times faster after an exact difflib-bound optimisation (cfb dev: 9 minutes to 12 seconds).
- Consequence: on natural errors a deterministic first stage cannot be the detector; used as a first-stage filter it would pass most wrong calls silently (detection 44% to 50%), which is exactly the "cheap verifier, large blind spot" failure of arXiv 2609.01345.
- This does **not** invalidate the injected-error result; it scopes it. It also does not say LLM judges are good here: Ma et al. report plain-input judge argument AUROC of 72.5 to 91.95 on fcrb and 63.4 to 78.9 on cfb (up to 95.1 on cfb with their derived-reference fix); we have not run any judge ourselves.

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4. Revised thesis and claim ladder

We no longer claim "a deterministic checker detects hallucinated tool calls". The study claims, in descending order of how much evidence is needed:

Claim	Evidence needed	Status
C1 (analysis). Grounding-based verification, deterministic or LLM-based, has a measurable ceiling set by the derivation path of each value; here is the per-path breakdown for checker and judges on public natural-error data.	DeRef (fcrb, cfb) + BFCL-Result, both systems per path	partly measured (Section 3); judges not yet run

Claim	Evidence needed	Status
C2 (system). A cascade whose first stage is deterministic **only for values it can ground** and whose second stage is an LLM that handles derivation (candidate-blind reference plus typed comparator) reaches accuracy within a margin of an LLM-only judge, with fewer LLM calls and a certified false-block budget.	judge panel, cascade simulation, Learn-then-Test	not started
C3 (guarantee). The false-block budget is stated correctly (marginal vs calibration-conditional), verified by repeated random splits, and its behaviour under dataset shift is measured.	re-splits, leave-one-dataset-out	not started
C4 (robustness). Under injected text in the context, LLM-judge verdicts flip more often than the deterministic grounding verdict, given source-trust labels.	AgentDojo runs, value-planting stress test	hypothesis
C5 (artefacts). Reproducible scripts, adjudication protocol, negative results.	repo	in progress

What we explicitly do **not** claim: first, novel mechanism, beats LLM judges, detects natural hallucinated arguments, immune to prompt injection, real agent errors (until natural-error sets are used and labelled).

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5. Hypotheses (revision of `review/PREREGISTRATION.md`)

Unchanged from the plan: H1 matched-FBR comparison, H2 cascade cost, H3 injection, H4 determinism, H5 independence, H6 coverage under shift. Changes and additions, to be logged in the pre-registration deviations before any judge run:

ID	Statement	Test	Change
H1	At a matched false-block rate, detection of S2 (deterministic) vs each judge on **natural** errors	cluster bootstrap with recalibration, one-sided non-inferiority margin 2 points, intersection-union across judges	now primary data = DeRef + BFCL-Result; **expectation: not supported for S2 alone**; the claim that matters is H1c below
H1c	The cascade S9 is non-inferior to the best judge at matched false-block rate	same	new
H2	Escalation share of the cascade <= 20% **at a stated error prevalence**, and **leakage** (errors the first stage passes that the judge would catch) reported	one-sided upper bound; Learn-then-Test over (tau_low, tau_high, judge cut)	adds prevalence and leakage (from Cheap Verifiers)
H6	Replace "FBR <= budget on 3 of 4 sets" (fails about half the time without any shift) by a **beta-binomial exchangeability null per dataset**, a KS shift diagnostic, and three repairs (recalibrate on 100/200/500 target genuine calls, weighted conformal, pooled calibration)	per-dataset upper-tail p-value inflated by the cluster design effect	new rule, fixed before any cross-dataset run
H7	Per derivation path (visible / format / default / history_only / no_source), judges and the deterministic checker differ in detection and false blocks as Ma et al. predict	per-path rates with CIs	new
H8	Role/slot-aware grounding recovers part of the *visible-but-wrong* errors	fcrb "visible" subset, paired comparison	new (add-on A4)
H9	Measured cost: tokens, mean/p95 latency per call per system on one machine class; judge-only vs cascade	descriptive + bootstrap	adds p50/p95; no dollars without prices you supply

Statistics (from Scout 3, to be implemented once): the primary guarantee is stated as **marginal over calibration draws** with a **calibration-conditional (PAC) headline**: with n genuine calibration calls and confidence 1-delta, pick the largest k' such that P(Bin(n, alpha) <= k'-1) <= delta (about 0.8 points lower threshold at 5% for n about 920); for several thresholds use Learn-then-Test with exact binomial p-values; calibrate the **end-to-end** cascade, not each tier; Mondrian strata with floors 1/(n_stratum+1) for risk tiers; report precision at realistic prevalence (at 5% budget and 1% error prevalence about 85% of blocks are false). Power: with about 838 errors per half a 2-point margin is powered at 80% only if systems disagree on at most about 5% of errors; a 1-point margin is not powered.

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6. Systems under test

Tag	System	Needs
S0	always allow / always block	nothing
S1	exact string match (our reimplementation, labelled so)	nothing
S2	ours: type-aware argument grounding (+ conformal threshold)	repo
S2b	S2 with source-typed grounding and role cues (add-ons A3, A4)	repo
S3	Saidso (deterministic grounding library)	`pip`/clone, README-level claims to be tested
S4	DeRef judge baselines from `SarielMa/DeRef_release` (CLM, Qwen3-8B/27B, Jev, Llama, Mistral)	GPU or OpenRouter; check each model's licence
S5	Groq/NVIDIA hosted judges, **frozen** call-level prompt (hash `9b59ae0a3fa5384a`) plus an argument-level prompt written before any run	provider keys
S6	ProvenanceGuard-style staged prompt on a small LLM	keys
S7	candidate-blind derived reference + typed comparator (DeRef idea with our normalisers)	keys/GPU
S8	**Cascade S2 -> S7 (or judge)** with Learn-then-Test thresholds	all of the above
S9	judge with k = 3 samples (score for conformalisation)	keys

A fairness rule: every system gets the same context text and the same tool schema; thresholds for all systems chosen by the same conformal rule on the same dev genuine calls; no system tuned on the test half.

7. Data plan

Role	Dataset	Access and licence (verified)	Use
Primary natural-error set, per-argument labels with derivation path	`lm2445/DeRef`: fcrb 1,260 calls, cfb 991 calls (long returns), fintrace 4,668 (rule-made), fcrb_relocate 1,370	open, Apache-2.0	H1, H1c, H2, H7; direct comparison with Ma et al.
Breadth of natural errors across 109 models	`HuanzhiMao/BFCL-Result` (2025-12-16 release)	open, Apache-2.0, 3.1 GB repo (sample files only)	natural errors without running models; needs adjudication
Real multi-turn trajectories with gold write actions	`sierra-research/tau-bench` `historical_trajectories`	MIT	provenance across tool returns
Human-labelled argument errors (small)	AgentHallu BFCL subset (36 "Incorrect Tool Arguments" + 61 clean)	CC-BY-4.0 per README	audit gold; tiny
Injection with real executed calls	AgentDojo `runs/` (`security=true` means the attacker call executed)	MIT	H3, value-planting test
Synthetic baseline	ArgGround-Bench (Glaive v2, LLM-generated)	Apache-2.0	scoped "injected errors" result only
Provenance experiments with real MCP outputs	MCP-Atlas	CC-BY-4.0	optional A9
Skip	Hallucinated-Tools Benchmark (no download found), xLAM (gated), AgentSuite mirrors (no licence declared; prefer upstream sources)		

Adjudication protocol (BFCL "wrong" often means "differs from gold", e.g. a Chinese place name copied from the user): every mismatched argument is labelled {ungrounded-wrong, grounded-but-wrong, format-equivalent, acceptable-alternative}. Step 1 automatic with independent rules (not our checker, to avoid circularity); step 2 an

LLM adjudicator using one of the judge models plus its disagreement flags; step 3 **you** audit a stratified sample of about 100 cases (45 to 60 minutes), reporting label precision with no inter-rater claim. DeRef's own authors report 3 to 4% unconfirmed labels on 100 audited items.

Splits: by conversation / `query_id`, 50/50, fixed seeds; DeRef dev already used exploratorily as above, test untouched; dev/test file hashes recorded.

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8. Novelty add-ons, ranked

Scoring: **Impact** on the professor's question and on the claim ladder; **Evidence** that it is open (scouts); cost in person-days; risk. Tier 1 = do for the core claim; Tier 2 = strong additions if time allows; Tier 3 = optional.

#	Add-on	Why it is worth it / new relative to what the scouts found	Cost	Risk	Tier
A1	**Judge panel at matched, certified false-block budgets** on DeRef + BFCL natural errors, with judge scores conformalised (log-prob, vote share, verbalised confidence)	Ma et al. report AUROC and an 80%-acceptance rule; ConformalGuard does this for email only. Answers the professor directly.	4 to 6	a judge wins everywhere (reported honestly)	1
A2	**Derivation-path analysis for the deterministic checker and judges** (visible / format / default / history_only / no_source), including the ceiling statistics	Ma et al. analyse judges only; a path breakdown for a deterministic checker turns weakness into the finding. First numbers already in Section 3.	2 to 3	low	1
A3	**Source-typed grounding + normaliser ablation:** admit schema defaults/enums and system policy as typed sources, exact-copy rules for opaque IDs, fix the 19% cfb "visible" misses; ablate each normaliser	Names the derivation gap without an LLM; ROPE names the cost of exact matching, nobody ablates typed normalisation	3 to 4	rules overfit a domain	1
A4	**Role/slot-aware grounding:** score a value against the argument's name/description and the nearest cue words and recency (my red-team: GBP injected as the *source* currency when the user asked for *to*-pounds; fcrb has 139 visible-but-wrong arguments)	Not found in the literature as a deterministic mechanism; tests whether the visible-but-wrong ceiling can be lowered	4 to 5	high: may not help	2
A5	**Certified cascade:** deterministic stage resolves only values it can ground; an LLM derives a candidate-blind reference for the rest; typed comparator and conformal budget; Learn-then-Test over (tau_low, tau_high, judge cut); report escalation, leakage and prevalence	Agent-Sentry/ToolFence cascade for injection, DeRef gives the derived reference; **an end-to-end false-block guarantee for argument grounding was not found**	4 to 6	LLM cost; derived references can themselves be wrong	1
A6	**Calibration-conditional headline + risk-tier Mondrian budgets + leave-one-dataset-out shift study** (fixed beta-binomial null)	corrects our guarantee wording; per-tool tiers with floors; shift evidence is thin in the literature	3	likely "does not hold under shift", still a finding	1
A7	**Independence and robustness:** phi coefficient between checker and each judge on natural errors; value-planting / injected-text stress test with and without a source-trust label on AgentDojo runs	Yang measured independence for shell actions; nobody for argument grounding; no head-to-head under injected text found	5 to 7	judges may barely move; trust labels depend on tool metadata	2
A8	**Allow / ask / block with two budgets** (false-block rate and interruption rate) in simulation	CORA/ProvenanceGuard have abstain/ask without both budgets; practical killer is false-block burden (a deployed repo reports a 40% ask rate "unusable")	3	needs a user-response model	3
A9	**Argument-origin labels from typed matching** (USER / TRUSTED_TOOL / UNTRUSTED_TOOL / INVENTED) replayed on AgentDojo and MCP-Atlas without an LLM	each provenance paper names LLM- or self-report-based provenance as its bottleneck; this would be a deterministic alternative	5	label accuracy on derived values	3

Recommended order if time is short: **A2, A1, A5, A6, A3**, then A7. The first four directly settle the professor's question; A3 repairs a concrete weakness seen in Section 3.

9. Work breakdown and schedule

Effort is one person part-time; compress as needed (minimum path below).

Phase	Days	Work	Outputs	Needs
P0 Freeze	1	write the pre-registration deviations (H1c, H6 rule, H7 to H9, derivation-path reporting), adopt the argument-level judge prompt, fix splits; correct our claims (Section 11)	updated `PREREGISTRATION.md`, README/STATUS/deck fixes	none
P1 Data	3	load DeRef (`fcrb`, `cfb`, `fintrace` via paging), BFCL-Result sample (about 20 diverse models, argument-level failures), tau-bench MIT trajectories; adjudication rules; investigate the 19% cfb misses	`benchmarks/load_deref.py`, `build_bfcl_natural.py`, data cards	disk (DeRef 32 MB; BFCL sample only)
P2 Baselines	2	S1, S2, S3 (Saidso) on all sets; per-path tables (A2)	per-path result JSON	none
P3 Pilot (gate G1)	2	two judges (one small, one large) on DeRef dev, matched FBR	pilot table	**provider keys**
P4 Judge panel	4	S4 to S6 on test halves, once each; conformalised scores; latency and tokens	judge results	keys; Kaggle GPU for local judges
P5 Cascade	4	S7 and S8 with Learn-then-Test; leakage and prevalence tables	cascade results	P4
P6 Guarantee and shift	3	calibration-conditional thresholds, re-splits (1,000), leave-one-dataset-out, repairs	coverage tables	P1 to P2
P7 Robustness and independence	5	AgentDojo replay, value planting, phi coefficient, prompt paraphrases, repeat-run variance	robustness tables	keys
P8 Write-up	4	tables/figures, deck, speaker guide, README/STATUS, handoff update	final documents	all

Gate G1 (after P3). If the large judge beats S2 and the cascade prototype cannot be within 2 points of it at matched false-block rate on dev, state the evaluation-study framing now and shorten P5 to the analysis. **Gate G2 (after P4).** If every judge dominates on all of H1c, H2, H3, H4, H5 the final claim is the negative result plus the benchmark and the derivation-path analysis (decision table in `RESEARCH_PLAN.md` section 8 stays valid).

Minimum viable path (about two weeks): P0, P1 for DeRef only, P2, P3, P4 with one open and one hosted judge, P5 simulated from saved scores, H1, H1c, H2, H7 only; state H3 to H6 as untested.

10. Compute, keys and tooling

- **Hosted judges:** Groq and NVIDIA OpenAI-compatible endpoints via `benchmarks/llm_judge_args.py` (written, self-tested, never run; prompt hash frozen). Needs `GROQ_API_KEY` / `NVIDIA_API_KEY` as environment variables; free-tier limits decide the seeded sample sizes (set `--limit` before running).
- **Local judges:** DeRef's repository scripts CLM, Qwen3, Llama, Mistral and Jev judges; local ones need a GPU (Kaggle T4, Internet On). Kaggle connector currently timed out/unauthenticated, so notebooks are delivered as files.
- **Disk:** `U:` about 2 GB free. Download only DeRef (32 MB) and individual BFCL result files; never clone the 3.1 GB repository.
- **Subagents:** use Sonnet; each scout hit the account's session limit once, so run at most three in parallel and ask each to write its file early.
- **Scripts to write:** `load_deref.py`, `build_bfcl_natural.py`, `analyze_paths.py` (A2), `llm_judge_args.py` extensions (temperature, per-argument output), `cascade_lt.py` (Learn-then-Test), `analyze_h1.py`,

`analyze_independence.py`, `make_injection_set.py`, `analyze_shift.py`.

11. Corrections to our existing materials (honesty debt)

Where	What is wrong or overstated	Fix
README, STATUS, deck slide 22, speaker guide	"Calibrated budgets held on held-out data" implies the method beat its guarantee; realised rates are below budget partly because of ties and chance (realised rate is Beta-distributed; about half of calibration draws exceed alpha)	state the guarantee as marginal under exchangeability, add the calibration-conditional version, remove "held" wording
Same	7.3% vs 14.0% presented without the natural-error result	scope to "injected absent-value substitutions in LLM-generated conversations" and add the Section 3 table
Deck methodology slide 18, speaker guide, handoff	"call score = weakest argument" and "traceable to context" presented as ours	credit Ma et al. (`min` aggregation) and ProvenanceGuard; reword the contribution as the measurement, the cascade and the guarantee
Research plan text	"novelty" statements	replace with the claim ladder in Section 4
Pre-registration	H6 rule invalid (fails about half the time under no shift)	add deviation entry (no cross-dataset run has happened yet)
`benchmarks/kaggle_full_run_summary.json`, README tables	none of the above are wrong; keep as is	no change
Handoff files	out of date after the scout	update `handoff/` in P0

12. Risks

Risk	Effect	Mitigation
Judges dominate on natural errors (likely)	deterministic stage cannot be the detector	cascade (A5) or evaluation-study framing; decision gates G1/G2
Cascade leaks errors the first stage passes (blind spot)	headline accuracy hides misses	report leakage and escalation per class and prevalence; derive tau_low conservatively
Label noise (BFCL "wrong" is not "ungrounded"; DeRef labels author-made)	misleading rates	adjudication protocol; per-path reporting; your 100-case audit; prediction-powered intervals
Exchangeability fails (clusters inside conversations; dataset shift)	guarantee violated	cluster-aware bootstrap, Dunn et al. style adjustments, leave-one-dataset-out, report violations
API limits, cost, privacy (benchmark text leaves the machine)	partial runs	seeded subsamples, cache all responses, open-model path
Fast-moving 2026 literature (several sources days old, unreviewed)	scooped or contradicted	re-check arXiv before submission; cite as preprints
Opus/Fable unavailable; subagent rate limits	slower scouting	Sonnet only, three in parallel, files written early

13a. Does choosing "evaluation-first" change the project or its intention?

It changes the **claim and the pitch**, not the **artefact or the goal**.

	Stays	Changes
Goal	a pre-execution gate for agent tool calls over MCP	nothing
Artefact	MCP server, proxy with risk tiers, schema validation, argument grounding, session tracking, all tests	grounding is presented as the cheap first stage of a cascade, not as "the detector"
Headline claim		from "our checker detects hallucinated tool calls" to "a calibrated cascade, with measured strengths and limits"
Review deck	required format and structure	contributions and results slides

Option C, "system first, evidence built in" (keeps your original identity): title and abstract stay about the gate ("a self-reflective pre-execution gate for MCP tool calls"); the evaluation is the proof section, and the claims are exactly those the data supports (cascade, guarantee, measured limits). Same work as option A, different emphasis. I recommend A or C, not B.

Why not keep the old claim: our own natural-error result and the closest paper mean a purely deterministic detector claim can be falsified by any LLM judge the professor runs. A measured claim cannot.

13. Decisions I need from you

1. **Framing (recommended: A or C).** Option A: "evaluation-first study with a calibrated cascade". Option B: keep the old system claim and present the natural-error result as a limitation (not recommended: falsifiable). Option C: system first, evidence built in (Section 13a). A and C differ only in emphasis.
2. **Provider keys** (`GROQ_API_KEY`, `NVIDIA_API_KEY`) set as environment variables, plus the free-tier limits shown on each dashboard, so P3 and P4 sample sizes can be fixed.
3. **Kaggle:** reconnect/authorise the connector and tell me your weekly GPU hours, or I deliver notebooks for you to run.
4. **Labelling:** the 100-case audit by you (default), or none with a stated caveat.
5. **Deadline and weekly hours** (chooses the two-week minimum path vs the full path).
6. **Which "modern LLM"** your professor means (a named model) so the strong-judge baseline matches.
7. **Permission to install and run Saidso** (an untrusted 1-star repository) in an isolated virtual environment as a baseline.

14a. What is already prepared (2026-10-20 evening)

Item	File	State
Key loader (environment, or a private file outside the repo)	`benchmarks/keys.py`	done; run `python benchmarks/keys.py` to see which keys are visible (names only)
DeRef loader with fixed dev/test split	`benchmarks/load_deref.py` -> `benchmarks/data/deref/cases_{fcrb,cfb}.json`	done (fcrb 630/630, cfb 507/484)
Deterministic baselines S1/S2 with per-path breakdown and a test-run ledger	`benchmarks/eval_deref.py`, `benchmarks/test_run_ledger.jsonl`	done
Judge script on DeRef (multi-call cases, context truncation option, `--allow-test` guard)	`benchmarks/llm_judge_args.py`	done, self-tested, never run against an API
Analyses H1 (matched false-block, cluster bootstrap), paths (H7), cascade with leakage	`benchmarks/analyze_study.py`	done, self-tested on synthetic data

Item	File	State
Bug fix: minus signs in number matching; faster fuzzy matching; 3 new tests	`server/pipeline/arg_ground.py`, `tests/test_arg_ground.py`	done, 31 tests pass
Pre-registration deviations, README/STATUS corrections	`review/PREREGISTRATION.md`, `README.md`, `STATUS.md`	done
Pilot runbook	`handoff/08_PILOT_RUNBOOK.md`	done

14. Next seven days (concrete)

1. (you) Decide Section 13 items 1 to 5; set provider keys.
2. (me) P0: update the pre-registration deviations, correct README/STATUS/deck/speaker guide wording (Section 11), refresh the handoff files.
3. (me) P1: write `load_deref.py`; investigate why 19% of cfb "visible" correct arguments are flagged (matcher limitation or label rule); build the BFCL natural-error sample.
4. (me) P2: run S1, S2, S3 on DeRef dev with per-path tables; ablate each normaliser (A3).
5. (me, after keys) P3: pilot two judges on DeRef dev, matched false-block rate; report Gate G1.
6. (you) Review the adjudication protocol and the 100-case sample once it exists.
7. (me) Commit the scout reports and the exploratory script to `review-eval-updates` only when you ask.

Appendix: verification status of the sources used in this plan

Verified by me against arXiv abstract pages (title, date, abstract) or the GitHub/Hugging Face APIs on 2026-10-20: 2610.18497, 2607.01236, 2610.07359, 2609.10969, 2512.03109, 2604.09155, 2608.10430, 2609.26550, 2605.11039, 2603.22868, 2609.37196, 2610.12292, 2610.11030, 2607.07405, 2608.26623, 2604.16706, 2608.27496, 2608.27146, 2512.06716, 2607.24343, 2610.09793, 2504.18333, 2503.00061, 2510.09023, 2504.11168, 2610.10742; repositories `KarthikRommula/Saidso`, `SarielMa/DeRef_release`, `HuanzhiMao/BFCL-Result`, `sierra-research/tau-bench`; datasets `lm2445/DeRef`, `AgentSuite/BFCL-trajectories`, `ibm-research/fc-reward-bench`. Not re-verified by me: the scouts' conformal-theory citations (Scout 3), the dataset field layouts beyond the two DeRef files I opened, and all numbers other than my own Section 3 measurements. Read depth per source is recorded in the scout reports.